Math 370
Summary Paper
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1. History and Preliminary Results

In "The Chromatic Number of the Plane is at Least 5 — A New Proof" [1], authors Geoffrey Exoo and Dan Ismailescu focus on tightening the lower bound for the Hadwiger-Nelson Problem. This problem seeks to answer:

> What is the smallest number of colors sufficient for coloring the plane in such a way that no two points of the same color are unit distance apart?

The problem was first formulated in 1950 by Edward Nelson. Within 10 years of its proposal, researchers were quickly able to determine some bounds for the solution to the problem, which is known as the chromatic number of the plane and denoted as $\chi(\mathbb{R}^2)$.

In 1961 a the lower bound of 4 was discovered by Leo and William Moser. They constructed a 7-vertex unit-distance graph that was later named the "Moser Spindle" (see Appendix, Figure 1). This graph had 7 vertices which were adjacent only if the distance between them is one. In this graph, each independent set has at most 2 vertices. Therefore, the graph requires 4 colors to be properly colored because there are 7 vertices in the graph. The motivation behind the Moser Spindle was that any lower bound on the chromatic number of a unit distance graph gives the same lower bound for $\chi(\mathbb{R}^2)$.

The upper bound of 7 was also proved in 1961, by Hugo Hadwiger himself (some give J. Isbell credit). The proof relies on a hexagonal tessellation of the plane (see Appendix, Figure 2). The diameters of the hexagon tiles are taken to be slightly less than unit distance and one color is assigned per hexagon. The plane is tiled in such a way that no two adjacent hexagons are given the same color. As seen in Figure 2, this configuration requires 7 colors. Furthermore, as can be easily verified, two points of the same color cannot be exactly one unit apart.

For close to 60 years, there had been no progress on these bounds. The goal for researchers in this field was to introduce a family of finite unit-distance graphs in the plane that are not 4-colorable. Finally, in 2018, the lower bound was raised to 5, by a non-mathematician, but rather a computer scientist and biologist named Aubrey de Grey. Although de Grey is credited for finding this result, Exoo and Ismailescu independently found another non-4-colorable finite unit-distance graph. Despite not being the first to publish this result, their construction (in addition to de Grey's) leverages a tool that was not available to Hadwiger, Nelson, or any of the mathematicians in the 60's: computer assistance.

2. Technical Preliminaries

Because this is a computer-assisted proof, the main technical preliminary is the algorithm the researchers use to determine that the graph is not 4-colorable. They use a modified Depth First Search (DFS) algorithm to arrive at their result. This algorithm is a recursive algorithm for searching all of the vertices. The steps of the algorithm are as follows:

(1) Enforce some ordering on the vertices
(2) Initialize every vertex to have all 4 colors available
(3) Pick a start vertex

(4) For each Color X in the list of available colors for the vertex:
(5) Color that vertex with Color X
    (a) Save state
(6) Visit all of the vertex's neighbors and remove Color X from their list of available colors
(7) Go to the next vertex in ordering and start from Step 4
(8) If we reach a coloring conflict (no colors are available for a particular vertex), return to the last saved state
    (a) If there is no saved state to return to, the graph is not 4-colorable

Because the DFS exhaustively checks all possible coloring combinations, its termination without having found a consistent coloring proves that there is no way to color the vertices of the given graph with the number of colors specified. This algorithm this can be thought of as an optimized brute-force methodology.

3. Main Result

The main result of this paper is the construction of a unit-distance graph which is not 4-colorable. At the time this paper was written, de Grey's paper had not yet been published and the researchers attempted to tighten the existing bound of $\chi(\mathbb{R}^2) \geq 4$. Exoo and Ismailescu's goal was to to make progress on this 60-year old problem.

They proved that $\chi(\mathbb{R}^2) \geq 5$ using a construction of a graph that, at each step, forced a desirable property in the subsequent graph. They then applied their DFS algorithm to determine whether a subgraph of their overall graph (and thus, by implication, the plane itself) was 4-colorable. This subgraph (G_{627}) was a unit-distance graph with 627 vertices, which can be seen in thhe Appendix, Figure 3.

The overall graph in which G_{627} was embedded stemmed from a separate construction. The authors created a graph with 18 equilateral triangles of side length $1/\sqrt{3}$, and proved that in any coloring of the plane, at least one of those triangles must be monochromatic. This property was verified with Sage, an open-source software package. They then placed a copy of G_{627} on each of the 18 equilateral triangles of side length $1/\sqrt{3}$ where the vertices that are colored red in Figure 3 coincide with the vertices of the triangles.

At this point, the authors were able to guarantee that at least one of the copies of G_{627} that is contained 18 times in their overall graph has these three monochromatic vertices. This completed their construction of the non-4-colorable finite unit-distance graph, G, which by my calculation is of order $3,625,314$. The only step that remained was to verify that this graph is not 4-colorable. To do so, the authors run the G_{627} through their DFS, with the initial condition that the three vertices of that central triangle share a color. When it terminates, the DFS outputs "No coloring". This concludes their proof. Note that G_{627} is a subgraph of G, and because G_{627} (with the property of the monochromatic vertices) is not 4-colorable, G is also not 4-colorable so $\chi(\mathbb{R}^2) \geq 5$.

There are still some questions I have on their constructive argument (some were asked during the presentation). These are:

(1) How were the authors able to guarantee that by adding copies of graphs on top of graphs, no additional edge would arise that would violate their prior verified claims?
(2) What was the intuition behind the authors' decision to use graphs which can be embedded in $\mathbb{Q}[\sqrt{3}, \sqrt{11}] \times \mathbb{Q}[\sqrt{3}, \sqrt{11}]$?

4. Future Work

At the end of the paper, the authors discuss many areas of future research work, specifically within the Polymath initiative, as well as other related research questions. They noted that following de Grey's breakthrough, a collaborative Polymath initiative emerged, dedicated to tackling the Hadwiger–Nelson problem. Within this undertaking, additional fields of research include finding increasingly smaller 5-chromatic unit-distance graphs and reducing reliance on computer-assisted proofs. Moreover, beyond these pursuits, the authors mention numerous compelling inquiries that remain wide open including: what is the minimum order of a 5-chromatic unit distance graph in the plane? Is it possible to use the newly discovered ideas to construct a unit distance graph which requires six colors?

Additionally, there is a lot of research centered around fractional chromatic numbers. The authors mention the research question related to proving that the fractional chromatic number of the plane is greater than 4. The authors discuss some results that have been found and note that recently Bellitto, Pěcher and Sédillo constructed a sequence of finite graphs whose fractional chromatic number approaches 3.8992.

5. Personal Reflection

While Exoo and Ismailescu's graph is of much larger order than de Grey, unlike de Grey, the first two claims of their proof were found to be manually provable without computer assistance. The authors stated that the reason the manual proofs were not included in the paper was becausee this finding developed after the submission of the report and the authors chose to maintain their original argument. The fact that their methodology does not rely on as much computer assistance might lead to advancements in this area of research of which a fully computer-assisted proof (that was not later found to be manually solvable) might not.

While some may think that the authors of this paper did not add anything to the discussion of the chromatic number of the plane, because they were not the first to introduce a new bound, I would disagree. As we have seen in our class, many of the results and famous theorems are related to ideas that were published that seemed trivial at first. I commend the authors for continuing their research and seeing it through even without officially getting credit as the "raisers" of the lower bound.

I thought it was interesting to see the depth-first search algorithm (one that we learn in DSA II) being used in practice. It is always interesting to see the interplay not only between two fields in real life, but even within two classes at Cooper. With the advent of computer assisted proofs, the unknown mysteries of graph theory can be resolved using computationally heavy methods that were previously thought to have been too complex to perform. The 60-year gap since the last known progress of the Hadwiger-Nelson problem is a testament to this fact.

6. Works Cited

[1] Exoo, G., Ismailescu, D. The Chromatic Number of the Plane is At Least 5: A New Proof. Discrete Comput Geom 64, 216–226 (2020). https://doi.org/10.1007/s00454-019-00058-1

7. Appendix

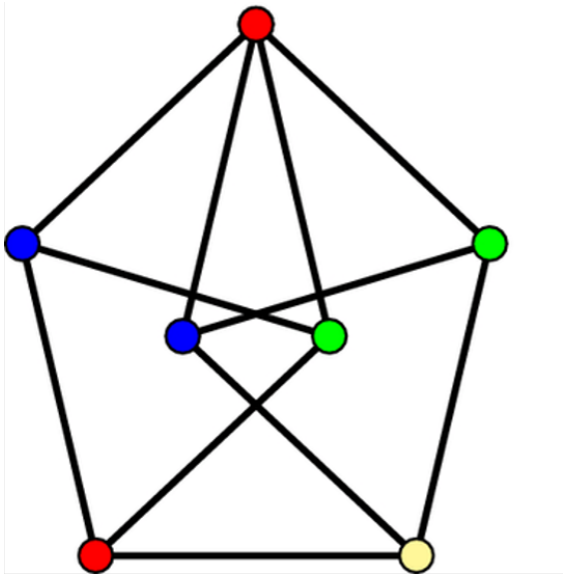

Figure 1. Moser Spindle

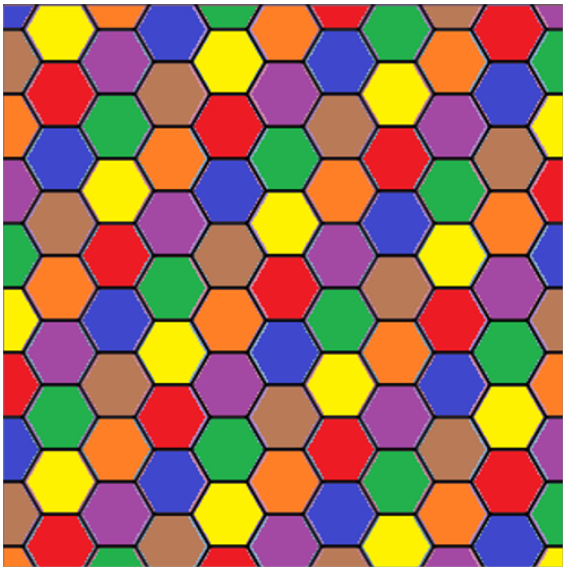

Figure 2. Hexagonal Tessellation

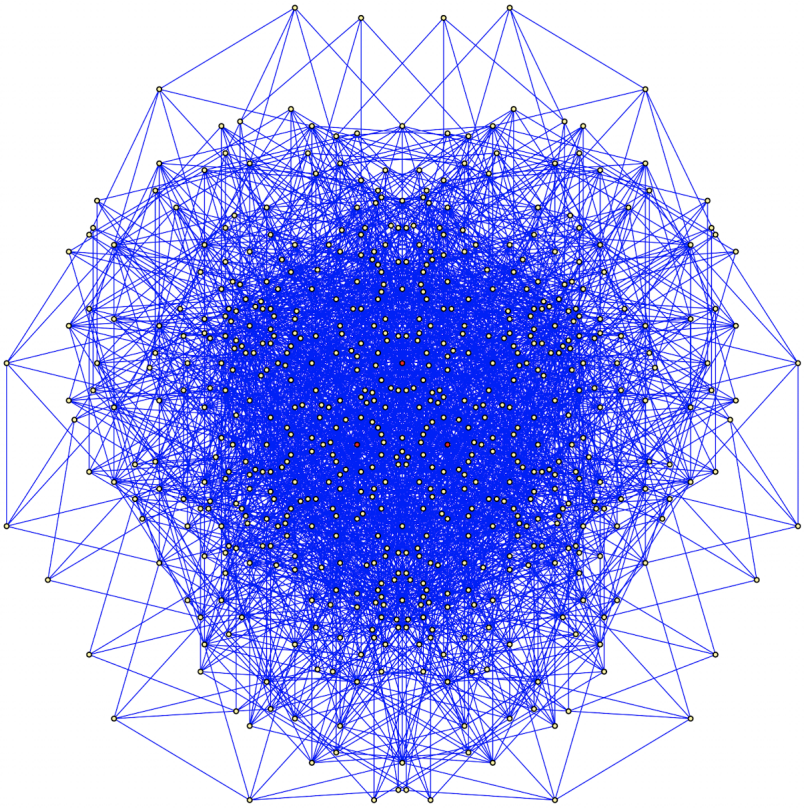

Figure 3. G_{627}